Team Project & GitHub

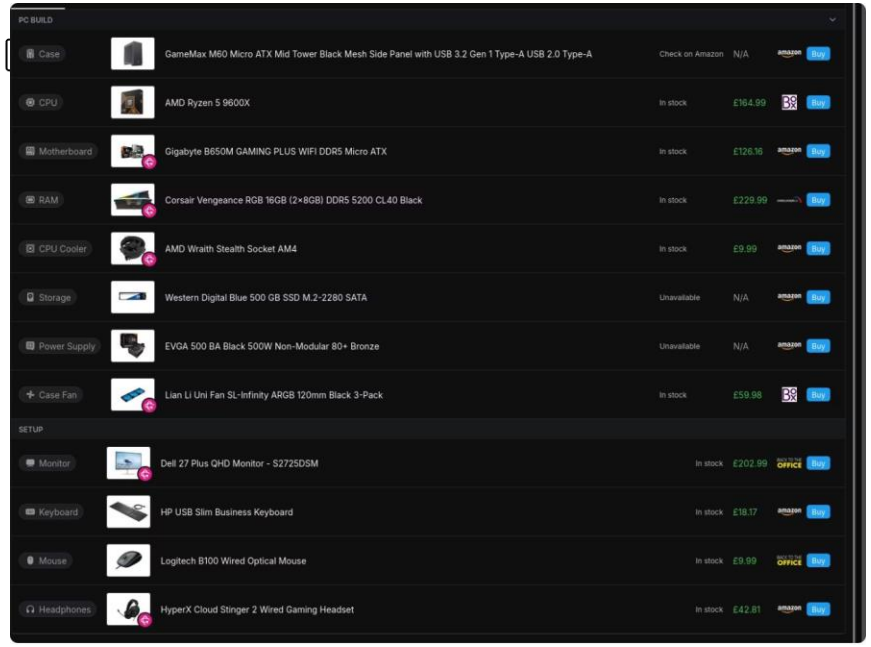
By Gowri Vinodkumar, Patryk Pospiech, Josh Webber, Saviour Oviawe



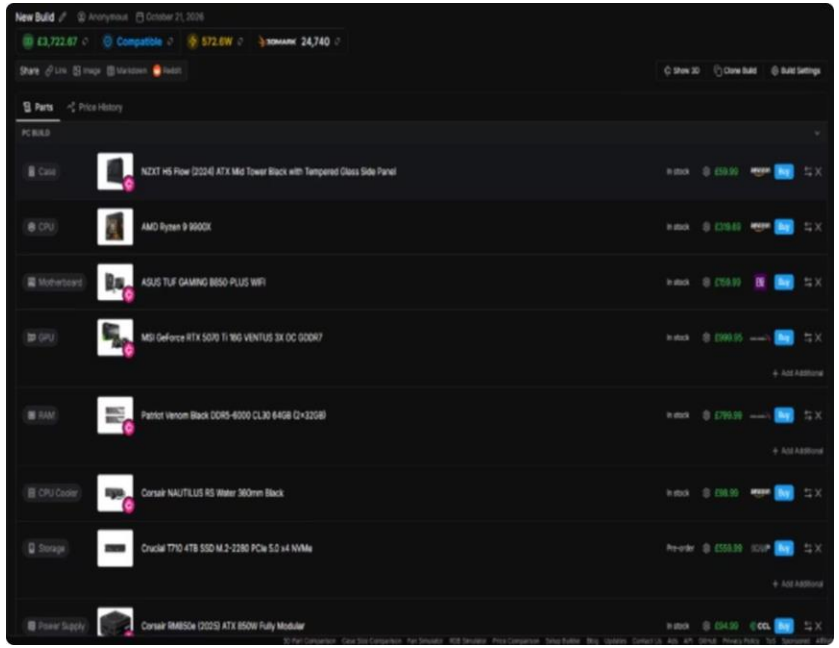
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Mini Project 1: Workstation Builds

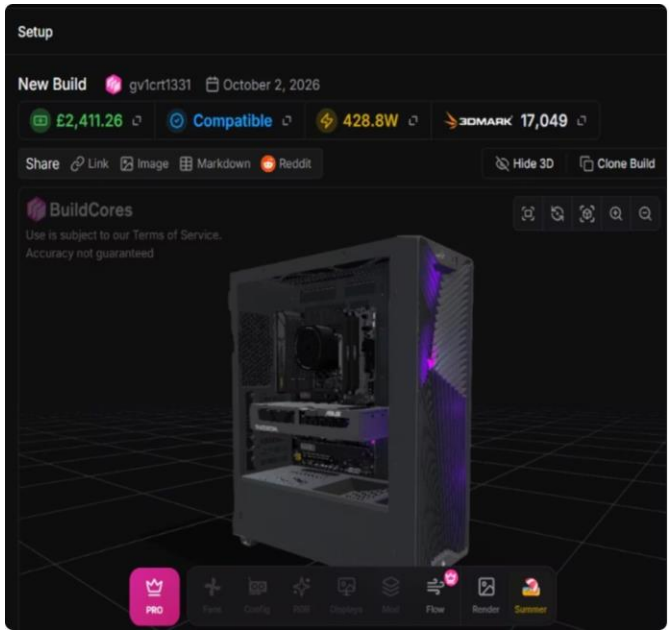
Project Manager



3D Visualisation Specialist Build



The CAD computer build



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Why these Components

We chose these components based on what each individual workstation needs to complete. Each workstation has a different job and workload so we needed to prioritize what was best for that workstation.



Admin Assistant

Basic, affordable hardware for emails, documents, spreadsheets and online meetings. The aim was to avoid spending money on unnecessary performance.



Project manager

A stronger office workstation with suitable memory, storage and display setup for reports, planning, reviewing drawings and client meetings.



Architect / CAD Designer

A stronger processor, dedicated graphics card and suitable memory for the 2d drawings, 3d models, texturing and rendering.



3D Visual Spec

High end workstation that focuses on performance, texturing and rendering capabilities

Overall aim: Match each workstation to the employee's needs, balance preformance with value for money and stay within the company's budget.

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Output:

employeeId	employeeName
1	Patryk
2	gowri
3	Josh

employeeRoleId	employeeId	roleDescription	roleBudget
1	1	Project Manager - Business Workstation	1250.00
2	2	Architect-CAD Workstation	2500.00
3	3	High-Performance Workstation	3750.00

workstationBuildId	employeeRoleId	workstationBuildRecommendation
1	1	Business workstation suitable for project management, office applications, multitasking, video conferencing and general producti
2	2	CAD workstation which includes high resolution monitors and large capacity storage that can store required programs and data, a
3	3	High-performance workstation suitable for demanding applications, graphics processing, multitasking and productivity across two

buildId	workstationBuildId	buildCpu	buildMotherboard	buildRam	buildStorage
1	1	AMD Ryzen 5 9600X	Gigabyte B650M DDR5 Micro ATX	16GB Corsair Vengeance RGB DDR5 5200	Western Digital Blue 500GB S
2	2	Intel Core i9-14900K	ASUS TUF GAMING B760-PLUS WIFI	Crucial Pro Overclocking 32GB DDR5 6000 CL36	Crucial P310 2TB M.2 PCIe 4.
3	3	AMD Ryzen 9 9900X	ASUS TUF GAMING B850-PLUS WIFI	Patriot Venom Black DDR5-6000 CL30 64GB (2x32GB)	Crucial T710 4TB M.2-2280 PC

totalBuildCost
£7,031.40

employeeRoleId	employeeId	roleDescription	roleBudget
1	1	Project Manager - Business Workstation	1250.00
2	2	Architect-CAD Workstation	2500.00
3	3	High-Performance Workstation	3750.00

roleDescription	roleBudget
Project Manager - Business Workstation	1250.00
Architect-CAD Workstation	2500.00
High-Performance Workstation	3750.00

roleDescription	roleBudget
Project Manager - Business Workstation	1250.00
Architect-CAD Workstation	2500.00

Client specification

What the pizza ordering system needs to do

CUSTOMER & ORDER INPUTS

Customer name, address and phone number

Order between 1 and 6 pizzas

Choose Small, Medium or Large for each pizza

Enter the number of extra toppings for each pizza

Choose whether delivery is required

PRICING & OUTPUTS

Pizza prices: Small £3.25 · Medium £5.50 · Large £7.15

Toppings: 1 = £0.75 · 2 = £1.35 · 3 = £2.00 · 4+ capped at 4

Apply 10% discount when subtotal is over £20, before delivery

Add £2.50 delivery charge when requested

Display subtotal, discount, delivery charge and final total

Core requirement: calculate the order accurately and show the customer a clear itemised bill.

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Variable specification: inputs

Variables store the information entered by the customer

Variable	Data type	Validation / constraint	Purpose
cName	String	1–50 characters; not blank	Customer name
cAddress	String	1–50 characters; not blank	Delivery address
pNumber	String	1–20 characters; not blank	Contact number
qtyPizza	Integer	1–6 pizzas	Number of pizzas
sizePizza	String array	Small, Medium or Large	Size of each pizza
qtyTopping	Integer array	Whole number, minimum 0	Toppings per pizza
delivery	Boolean / choice	Yes or No	Whether delivery is needed

Arrays link each pizza's size and topping count to the same order item.

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Variable specification: processing & output

Calculated variables turn the order inputs into a final bill

subTotal

PROCESS

Stores pizza and topping costs before discount and delivery.

discount

PROCESS

Stores the 10% discount when the subtotal is over £20.

total

OUTPUT

Final amount after discount and any delivery charge.

Calculation order: subtotal → discount (if eligible) → delivery fee → final total

The original specification defines all three as non-negative calculated amounts.

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Pseudocode: how the algorithm works

The pseudocode turns the client requirements into a sequence of instructions

01 INPUT

Ask for customer details and validate them. Name and address must be 1–50 characters; phone number must be 1–20 characters.

02 REPEAT

Validate the pizza quantity (1–6). Use a FOR loop to collect the size and topping count for every pizza.

03 PROCESS

Add each pizza's price and topping charge to the subtotal. Apply the 10% discount only when the subtotal is over £20.

04 OUTPUT

Add £2.50 if delivery is requested, then display the order details, subtotal, discount and final cost.

Why this structure? Validation catches unsuitable input; loops avoid repeating the same instructions for each pizza.

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Pseudocode: design justification

Why these decisions were made in the original project

DATA VALIDATION

The pseudocode checks input before continuing.

Invalid entries are sent back to the question instead of being accepted.

This reduces input errors and helps prevent the program from failing.

Customer details have length limits, while size and delivery use allowed choices.

TOPPING LIMIT & ROBUSTNESS

The written justification says toppings are capped at four.

This avoids needing to handle unlimited topping counts.

The justification explains this choice as supporting future changes, such as storing orders in a database.

A consistent cap also keeps topping-price calculations straightforward.

The main design aim was to make the program easier to use, more reliable and simpler to extend.

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Build table: implementing the design

Each stage of the build corresponds to a step in the pseudocode

STAGE	BUILD / PURPOSE
Menu	Display pizza sizes, prices and topping charges so the customer can compare costs.
Customer details	Collect and store name and phone number; ask for an address when delivery is required.
Quantity	Accept a whole number from 1 to 6 and repeat the prompt if it is invalid.
Pizza loop	For each pizza, ask for a valid size and topping count, then store both in lists.
Delivery	Ask for Yes or No and set the delivery charge when the customer chooses Yes.
Bill	Calculate subtotal, apply eligible discount, add delivery and show the final order details.

The build follows the algorithm so each input is collected before the final calculation.

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Build table: calculations & output

The final stage applies the pricing rules and produces the order summary

1. PIZZA COST

Add £3.25 / £5.50 / £7.15

2. TOPPINGS

Add the charge for each pizza

3. SUBTOTAL

Combine pizza + topping costs

4. DISCOUNT & DELIVERY

10% discount if subtotal > £20; add £2.50 delivery when requested.

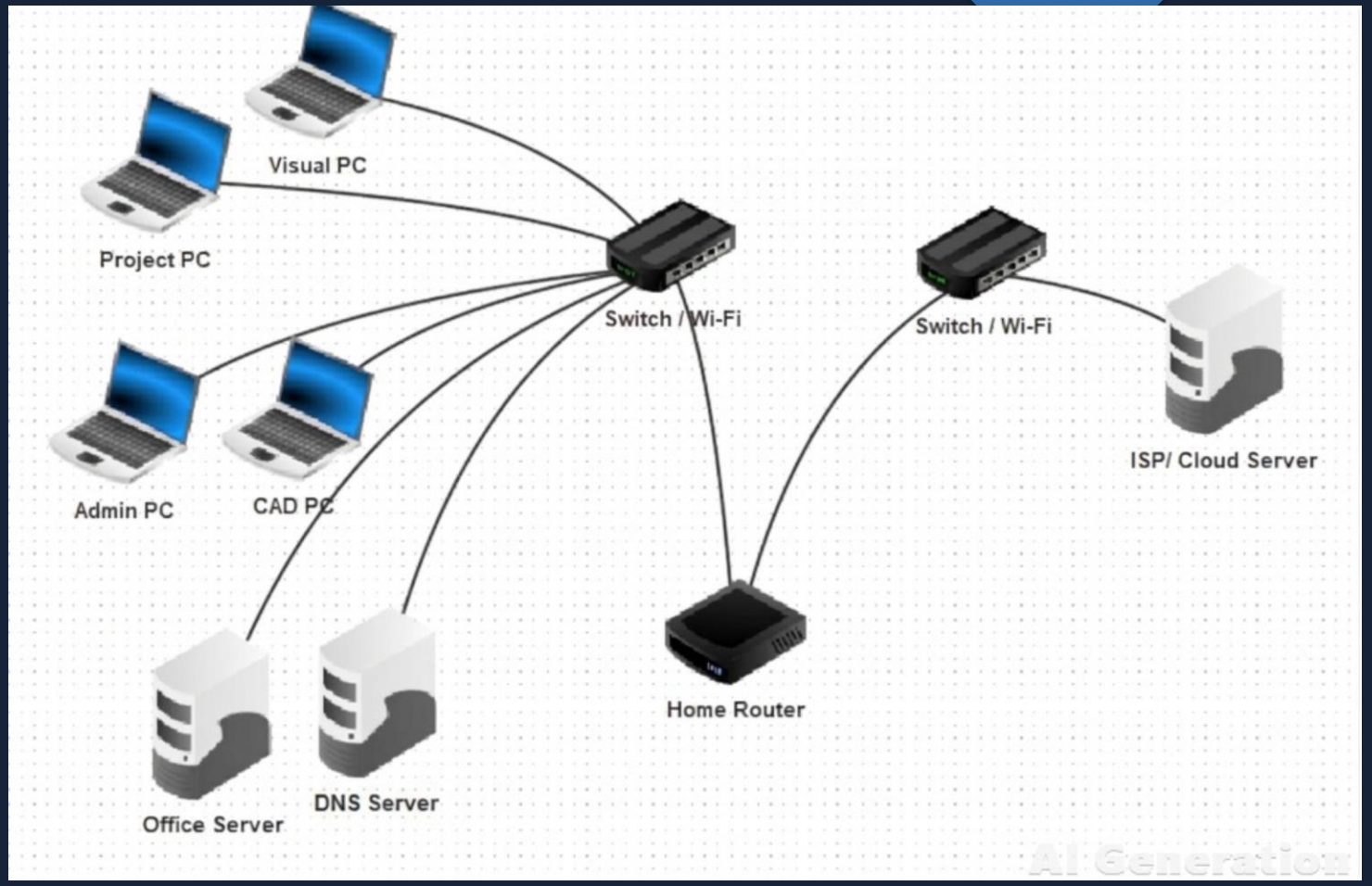
5. DISPLAY THE BILL

The build section shows how the algorithm's decisions become a working order and final cost.

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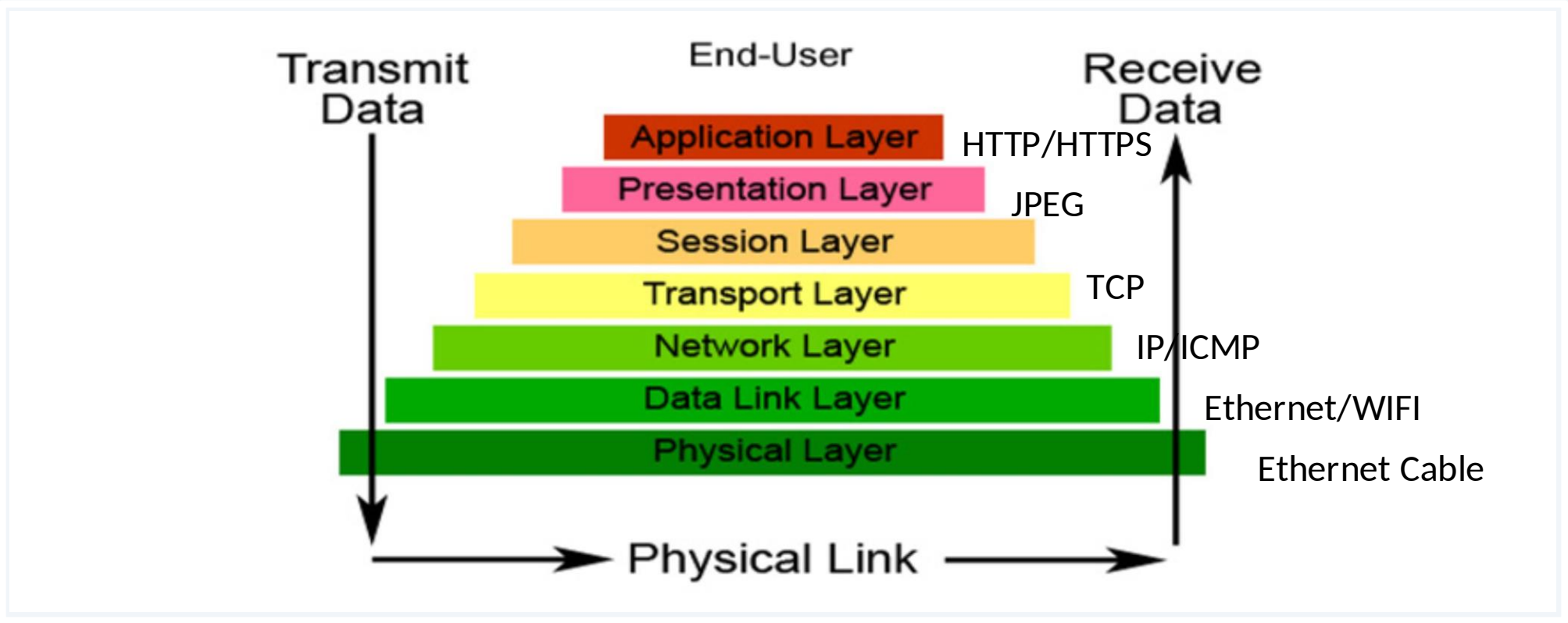
Project 3 : **NorthWest Architecture Studio Build**

- Model Working Office LAN and connection to external cloud network
- Test network
- Principles



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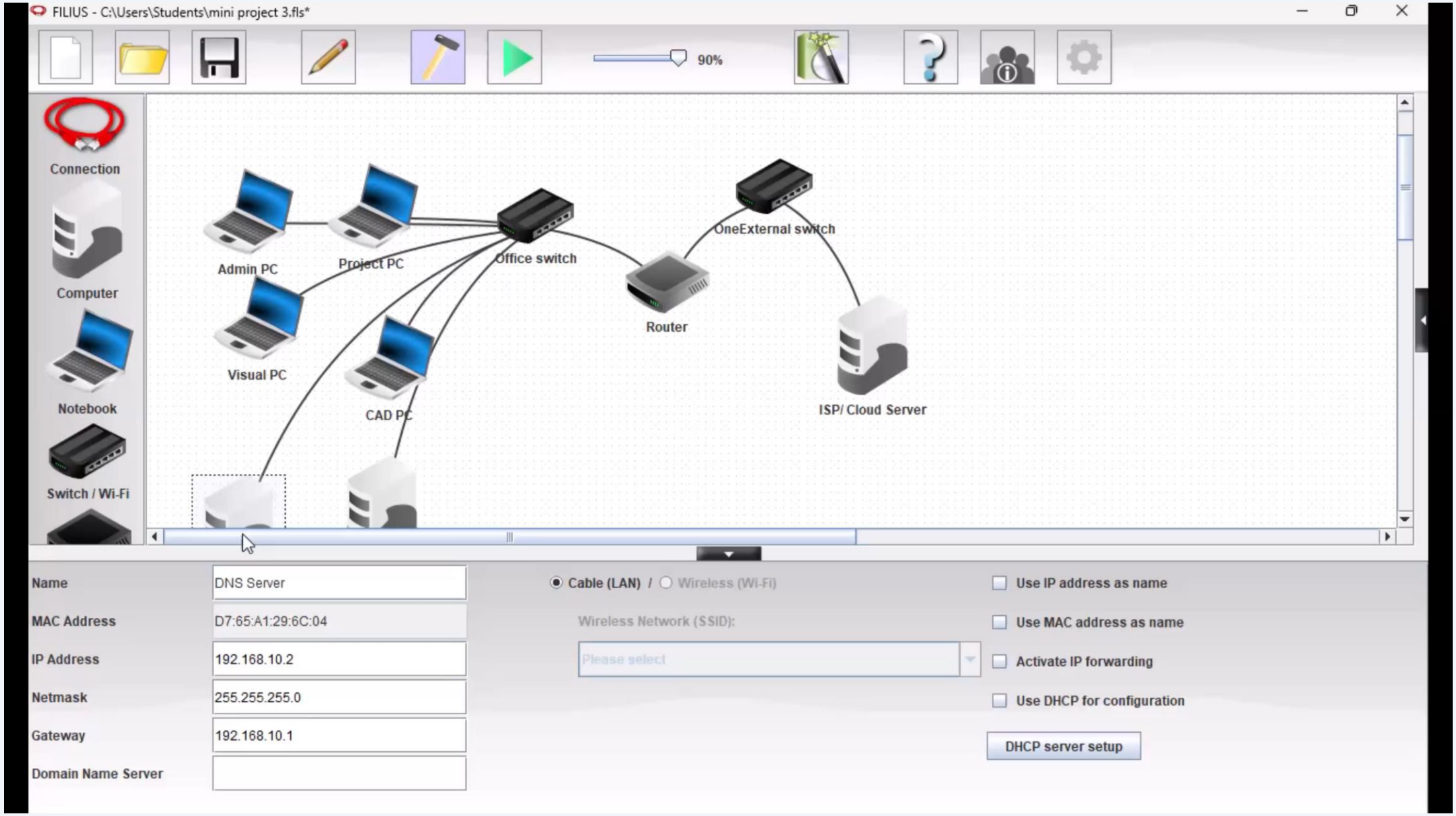
Layers and Protocol



Each layer has a specific job, from transmitting electrical or wireless signals to providing network services to applications.

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Main project: what the system must do



Cyber security Awareness

- Password Hygiene
- Multi factor authentication

- Phishing and social engineering
- Safe use of device
- Safe use of public Wi-Fi

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PASSWORD HYGIENE

Strong passwords protect accounts

Password hygiene is how users manage passwords and secure accounts from malware and hackers.



8+

At least 8 characters, with multiple character types



Unpredictable and random; length and number of characters matter



Computer-generated passwords are more random than human-made ones



Don't share passwords; use multifactor authentication

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PASSWORD HYGIENE

Password managers and reuse



13

Password managers help users make strong passwords, but a study of 13 managers found some generated predictable patterns



Use reputed managers and keep software updated; scammers build fake managers that can lead to data breaches



Credential reuse: one breach exposes all accounts that share the same login



Credential stuffing: hackers try stolen credentials on other sites

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TWO-FACTOR AUTHENTICATION

Passwords alone are weak

People use pets, family, friends, food and places in passwords, which makes guessing easier.



96%

Google: 2FA prevents bulk phishing attacks



76%

Google: 2FA prevents targeted attacks

- 2FA requires two isolated verification factors

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TWO-FACTOR AUTHENTICATION

Three categories of factors



Knowledge

Something you know, such as a password (the weakest link)



Possession

A mobile phone, security token or hardware fob; one-time passwords are valid for ordinarily 45-60 seconds



Inherence

Biometrics such as fingerprint or facial recognition

SMS to a phone number is vulnerable to SIM cloning (ENISA)

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MULTI-FACTOR AUTHENTICATION

MFA combines two or more factors

0.0079%

Compromise rate

1.44%

Success rate per attempt, even when the attacker knows the login credentials (arXiv)



Username and password



Two or more factors



MFA

Adds maximum attempts with temporary lockout, and behaviour factors such as typing dynamics for an adaptive policy

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PHISHING

What is phishing?

A social engineering attack: messages disguised as legitimate brands trick victims into revealing information or clicking malicious links.



Bulk phishing

generic emails to many people



Spear phishing and whaling

personalised, targeted messages



Pharming

redirects to fake websites that capture credentials



Smishing and vishing

text or voice message attacks



AI-powered phishing

convincing messages that evade detection

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PHISHING

How phishing works

Spoofing identity

A forged sender or impersonated brand



Social engineering hook

Urgency, fear or curiosity



Lure engagement

Click a link, open an attachment or enter credentials on a fake site



Harvesting and exploitation

Unauthorised access, financial fraud, identity theft or malware

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PHISHING

Phishing in the real world



Half

of UK businesses had a cyber attack in the last year, phishing the most common



32%

of charities had a cyber attack in the last year



7.78 million

cyber crimes affected UK businesses

AVERAGE COST OF THE MOST DISRUPTIVE ATTACK

£6,940

for businesses

£40,400

for medium and large firms

Ipsos, Cyber Security Breaches Survey 2024

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SAFE USE OF PUBLIC WI-FI

Public Wi-Fi can be unsafe

Common in cafes, airports and libraries, but attackers can access the network and steal data.

Threats

- Data interception of passwords and bank details
- Fake 'evil twin' networks
- Malware spread through unsecured networks

Avoid

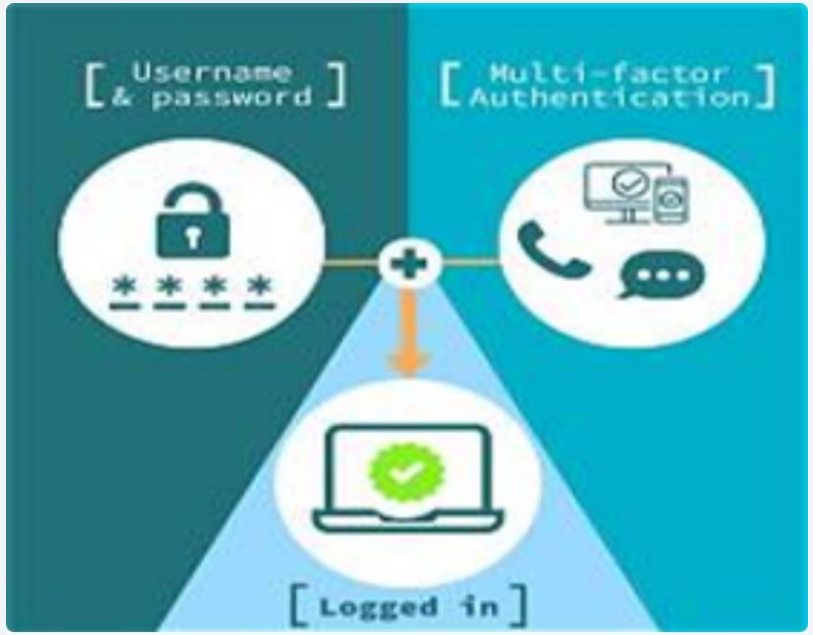
- Don't enter personal information
- Use mobile data for banking and other important tasks
- Turn off automatic Wi-Fi connection

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Emails: main source of phishing



Use reputed password manager

- Multi factor authentication & Unauthorized access
- Password strength

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The intended result described in the setup guide

PRIVATE

Only the invited team and tutors can access the portfolio.

SHARED

All invited members can edit the same repository and contribute to the work.

ORGANISED

The repository starts from the course template, giving the team a consistent folder structure.

If someone cannot see the repository, check whether they accepted the email invitation.

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Thank you

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